GeoPhysicsLandslideNet: A Physics-Closed Pentastream Architecture for Multimodal Landslide Segmentation

Samreedh Bhuyan[1]* Dr. Anil Earnest[2]

[1]School of Computer Engineering, Kalinga Institute of Industrial Technology (KIIT), India

[2]CSIR Fourth Paradigm Institute (CSIR-4PI), Bengaluru, India

July 17, 2026

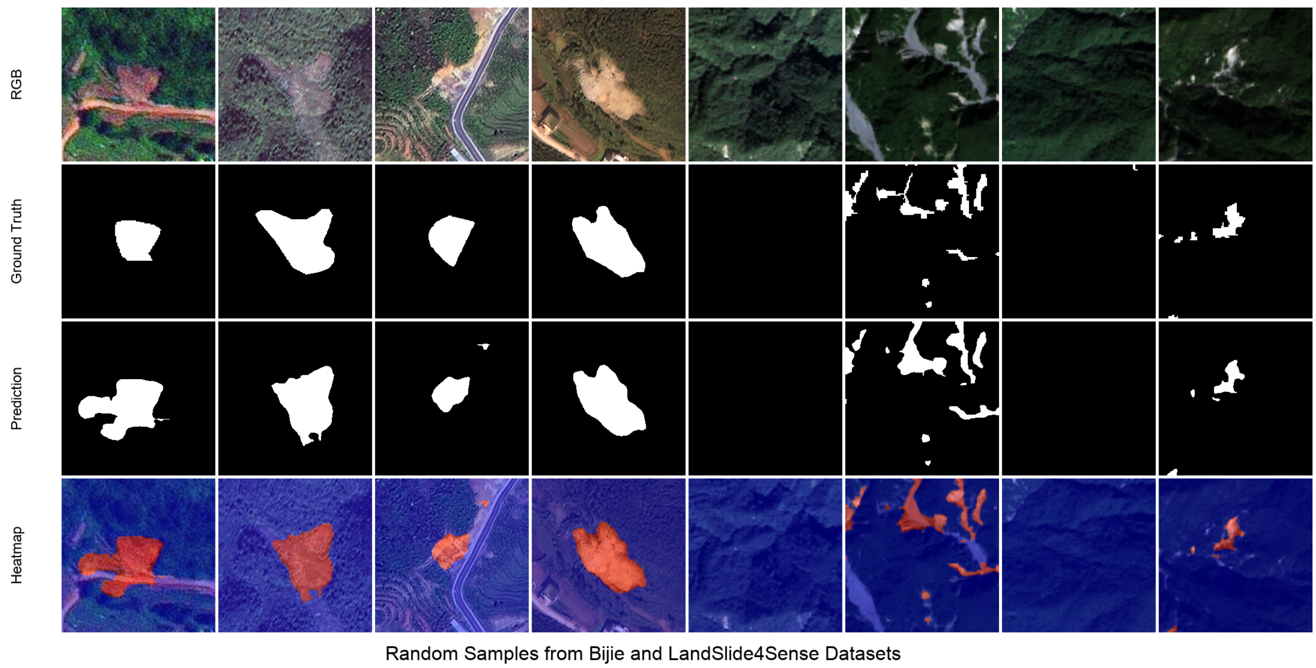


Figure 1: Qualitative segmentation examples on randomly selected validation samples (RGB, ground-truth mask, predicted mask, and probability heatmap overlay). Red denotes landslide confidence; blue denotes background.

Abstract

Landslide inventory mapping from remote sensing remains brittle when models rely on purely statistical fusion of optical and topographic cues. We introduce **GeoPhysicsLandslideNet** (PS-GPLNet), a pentastream segmentation architecture in which the same Taylor-stabilized infinite-slope factor-of-safety (FS) formalism appears at the encoder inlet, fusion core, skip pathway, decoder cells, and dual-path output routing. Five parallel encoders—EfficientNet RGB/DEM, physics encoders on each modality, and Prithvi-EO with LoRA—are hybridized by Complementary Modality Bridges, fused by MAO-GeoEGCA and TTEB, decoded by dual physics decoders with PGDI, and merged by Mechanistic Path Equilibrium Fusion (MPEF). On Bijie, PS-GPLNet attains best-epoch validation $\mathrm{F1} = 0.933$ and $\mathrm{IoU} = 0.907$, exceeding strong CNN and dual-stream baselines under a unified protocol. On Landslide4Sense it reaches $\mathrm{F1} = 0.736$ and $\mathrm{IoU} = 0.660$. We argue that closing the geotechnical loop inside the network—rather than treating physics as a post-hoc regularizer—is a distinct and competitive inductive bias for landslide segmentation.

1 Introduction

Landslides threaten lives and infrastructure across mountainous terrain. Optical remote sensing and digital elevation models (DEMs) enable automated mapping, yet pixel-level segmentation remains difficult: scarps, debris, and disturbed soils share spectral signatures with dry riverbeds and construction, while landslide polygons vary strongly across biomes [1, 2, 3].

Deep segmentation—from U-Net [4] and DeepLabV3+ [5] to transformer hybrids [6, 7, 8]—has replaced shallow classifiers for dense masks, and multimodal RGB+DEM streams consistently outperform single-modality models on benchmark tiles [3, 2, 9]. Dual-stream gated designs such as DiGATe-style networks learn to weigh spectral versus topographic evidence [10], but fusion and decoding remain statistically defined: gates are free parameters, not statements of slope stability.

Physics-informed geo-AI has advanced *susceptibility* mapping with infinite-slope factor of safety (FS), Newmark displacement, or hydromechanical priors in losses and sampling [11, 12, 13, 14, 15, 16, 17]. Pixel-accurate *segmentation* still seldom embeds the same mechanics as a first-class architectural primitive spanning encoding, fusion, and decoding. We therefore ask whether a single closed-loop FS inductive bias can organize a multimodal segmentation network end-to-end. PS-GPLNet is our answer: five parallel encoders hybridized by Complementary Modality Bridges (CMB), fused by MAO-GeoEGCA and TTEB, decoded by dual physics pathways with PGDI, and merged by Mechanistic Path Equilibrium Fusion (MPEF) driven by relative failure energy rather than scalar learned gates. Under a unified training protocol on Bijie [3] and Landslide4Sense [2], PS-GPLNet reaches best-epoch validation F1 = 0.933 / IoU = 0.907 on Bijie and F1 = 0.736 / IoU = 0.660 on Landslide4Sense, improving over strong CNN, transformer, and dual-stream baselines in our harness.

2 Related Work

2.1 From inventories to dense landslide mapping

Regional landslide studies historically combined field inventories with heuristic GIS overlays on slope, curvature, and hydrological indices [18, 19]. With open high-resolution optical stacks and DEMs, the community shifted toward *dense* polygon mapping: each pixel is labeled landslide or background, enabling direct comparison with building footprints, roads, and runout models. Bijie [3] and Landslide4Sense [2] crystallized this trend with public train/validation splits and reported CNN baselines. The task is imbalanced (scar pixels are rare), spectrally ambiguous, and topographically structured—properties that motivate both multimodal fusion and mechanistic priors.

2.2 Convolutional and transformer segmenters

Encoder–decoder CNNs remain the workhorse for geospatial segmentation. U-Net [4] and FCN [20] established skip-connected decoding; DeepLabV3+ [5] added atrous spatial pyramid pooling for multi-scale context; LinkNet and DEP-UNet variants in our ablation ladder inherit the same inductive bias. Transformers [21, 8] brought global context: Swin [22], SegFormer [6], and TransUNet [7] appear frequently in remote sensing leaderboards. These models excel at appearance and long-range dependencies but treat DEM derivatives as extra channels rather than as constraints from slope stability theory. PS-GPLNet retains EfficientNet [23] towers for representation learning while routing topographic proxies through parallel physics encoders.

2.3 Multimodal optical–topographic fusion

Fusing optical imagery with DEM, slope, aspect, or curvature is now standard for landslide detection [1, 3, 2]. Early fusion stacks channels at the input; mid-level fusion merges feature pyramids. BiFusion-Net [9] and related multi-stream CNNs demonstrate that DEM-guided features reduce false positives on uniform steep slopes. Our Complementary Modality Bridge differs by *explicitly* pairing each CNN level with FS-gated physics features before tri-stream attention, so appearance and mechanistic maps are aligned per scale rather than only at the bottleneck.

2.4 Dual-stream and gated architectures

Dual-stream U-Nets separate RGB and DEM encoders and reunite them at the decoder [10]. DiGATe [10] adds gated attention between streams and an adaptive decoder. We include dual_stream_gated and Dual-Stream UNet as *empirical baselines only*. PS-GPLNet is not organized around learned scalar gates: path weights at the output arise from softmax failure energies (MPEF) computed from the same FS cell family used in the encoder inlet. Shared conventions—two geospatial inputs, deep supervision, Tversky or Dice losses [24, 25]—are field practice, not evidence of architectural equivalence.

2.5 Geotechnical susceptibility and slope stability

Infinite-slope FS compares resisting and driving forces on a potential slip surface [12]. Newmark [11] extends the framework to earthquake-induced displacements. These models underpin landslide susceptibility zoning when soil parameters are assigned from geology or calibrated statistically. They are interpretable and physically grounded but static: they do not, by themselves, produce segmentation masks at Sentinel or aerial resolution, nor do they ingest foundation-model embeddings. PS-GPLNet imports the FS *structure* as differentiable cells and gating signals, not as a post-classification threshold on a susceptibility raster.

2.6 Physics-informed and physics-guided learning

PINNs and physics-guided ML inject PDE residuals or FS inequalities into training [16, 13, 14, 15, 17]. Most geo-AI physics work targets susceptibility maps or point-wise failure probability, with physics entering through the loss or auxiliary heads. Differentiable rendering of geotechnical parameters has improved stability in recent susceptibility networks [13]. PS-GPLNet instead closes the loop *inside* the forward pass: bounded material parameters, pixel/latent mechanistic cells, PGDI skip injection, and MPEF routing all share one FS skeleton. Physics regularizes representation and routing, not only the final scalar objective.

2.7 Earth-observation foundation models

Large-scale pretraining on multi-temporal EO stacks—exemplified by Prithvi-EO [26, 27, 28]—provides transferable context for downstream segmentation and change detection. Full fine-tuning is costly; LoRA [29] adapts frozen backbones efficiently. We treat Prithvi as a fifth stream projected to the fusion width, then fuse with MAO-GeoEGCA against CMB-hybridized RGB/DEM features. To our knowledge, few landslide segmenters combine foundation-model context with dual mechanistic decode paths and FS-closed routing.

2.8 Positioning

Table 1 summarizes how PS-GPLNet relates to representative lines of work. The novelty claim is architectural closure: the same FS formalism at inlet, fusion, skips, decoder cells, and output equilibrium—implemented in `from_first_steps/` and documented in `model_architecture.md`.

3 Method

3.1 Problem and notation

Given streams $x_1 \in \mathbb{R}^{B\times 3\times H\times W}$ (RGB) and $x_2 \in \mathbb{R}^{B\times 3\times H\times W}$ (NDVI+slope+DEM on Landslide4Sense; replicated DEM on Bijie), predict a binary mask $y \in \{0,1\}^{B\times 1\times H\times W}$. Proxies (α, h, m)—slope angle, elevation/soil-column proxy, moisture/vegetation—are derived inside the forward pass.

The infinite-slope factor of safety is

$$\mathrm{FS} = \frac{c + \phi\, h \cos^2\alpha}{\gamma\, h \sin\alpha\cos\alpha + w_m m + \varepsilon}, \tag{1}$$

with positive material parameters obtained via bounded exponentials $c = \exp(\mathrm{clip}(w_c))$ (and likewise for ϕ, γ, w_m) for numerical stability. Failure energy $\mathrm{FE} = \mathrm{ReLU}(1 - \mathrm{FS})$ drives gating and path routing.

3.2 Pentastream encoding

Five encoders run in parallel: EfficientNet [23] RGB/DEM towers $E_{\mathrm{rgb}}, E_{\mathrm{dem}}$; PhysicsEncoders $P_{\mathrm{rgb}}, P_{\mathrm{dem}}$ with pixel/latent mechanistic cells; Prithvi+LoRA T_{fm}. At each EfficientNet level i, Complementary Modality Bridge (CMB) merges CNN features with spatially aligned physics features into hybrid streams $H^i_{\mathrm{rgb}}, H^i_{\mathrm{dem}}$.

3.3 Fusion: MAO-GeoEGCA and TTEB

At neck/bottleneck levels, MAO-GeoEGCA performs gated cross-attention between Prithvi and hybrid RGB/DEM maps. TTEB builds skip tensors by mixing neighbouring pyramid levels across three streams with a softplus stability residual $\delta = |\psi - R/(D + \varepsilon)|$.

3.4 Dual physics decoder and MPEF

Two PhysicsDecoders share fused context but use stream-specific (α, h, m) and bottleneck residuals. PGDI injects TTEB skips through a learned β gate on a latent mechanistic cell. Path logits are merged by MPEF:

$$[w_A, w_B] = \mathrm{softmax}([\mathrm{FE}_A, \mathrm{FE}_B]), \quad O = w_A O_A + w_B O_B + \mathrm{Conv}_{1\times 1}([(\tag{2}$$

with entropy regularizer $\mathcal{R}_{\mathrm{MPEF}} = -\mathbb{E}[w_A \log w_A + w_B \log w_B]$.

Forward sketch. (1) Derive (α, h, m); encode five streams. (2) CMB hybrids; align Prithvi. (3) MAO at levels $2, 3$; TTEB skips at $0..3$. (4) Dual PhysicsDecoder $\to (O_A, O_B)$. (5) MPEF $\to$ (main, aux2, aux3, reg).

3.5 Closed-loop inductive bias

The pixel cell $g_{\mathrm{pix}}(x; \alpha, h, m) = f(x) \odot \sigma(\psi - \mathrm{FS}(\alpha, h, m))$ and MPEF weights $w = \mathrm{softmax}(\mathrm{FE})$ share the same FS skeleton, so proxies gate both features and path routing. Bounded $\exp(\mathrm{clip}(w))$ keeps FS finite during training.

3.6 Objective

Deeply supervised Tversky loss [24] on main/aux heads with weights $(1.0, 0.6, 0.4)$ plus $\lambda_{\mathrm{reg}} \sum_h \mathcal{R}^{(h)}_{\mathrm{MPEF}}$.

4 Experiments

4.1 Datasets and protocol

Bijie [3]: high-resolution optical+DEM landslide/non-landslide tiles. **Landslide4Sense** [2]: multi-region Sentinel patches with RGB–NDVI–slope–DEM. Inputs resized to 256×256. Optimizer Adam ($3 \cdot 10^{-4}$, weight decay 10^{-4}), Tversky α=0.3, β=0.7 on L4S and α=0.7, β=0.3 on Bijie, batch size up to 32 (Bijie batch 2 in our final run), 100 epochs, threshold 0.5. EfficientNet-B4 backbones frozen when pretrained; Prithvi frozen with LoRA. Metrics: accuracy, precision, recall, F1, IoU on the main head.

4.2 Baselines

We compare against LinkNet, DEP-UNet, GMNet, RMAU-Net, TransUNet, EMR-HRNet, ShapeFormer, Dual-Stream UNet, U-Net, DeepLabV3+, and dual_stream_gated under the same ablation harness. TriEncoderCFMNet is excluded (unpublished). PS-GPLNet numbers are best validation epoch from `outputs_final_bijie` and `outputs_final_l4s`.

Table 1: Conceptual positioning (dense segmentation setting).

Approach	Dense mask	FS in forward	Foundation context
CNN / transformer U-Net	✓	–	optional
Dual-stream gated [10]	✓	–	rare
Susceptibility PINN / PGML	often raster	loss / head	rare
PS-GPLNet (ours)	✓	✓	Prithvi+LoRA

4.3 Quantitative results

Table 2 and Table 3 report best-epoch validation scores. On Bijie, PS-GPLNet reaches F1 = 0.933 and IoU = 0.907, surpassing DeepLabV3+ (0.927/0.899) and dual_stream_gated (0.897/0.869). On Landslide4Sense, PS-GPLNet achieves F1 = 0.736 and IoU = 0.660, ahead of DeepLabV3+ (0.710/0.636) and the dual-stream variants in our ladder.

Figure 2 and Figure 3 visualize the F1/IoU spread across all harness models; the IoU ranking charts in Figure 4 and Figure 5 highlight that PS-GPLNet tops the L4S ladder and matches or exceeds the strongest CNN on Bijie. Heatmaps in Figure 6 and Figure 7 show precision–recall–F1–IoU jointly, where our model trades slightly lower recall on Bijie for higher precision relative to DeepLabV3+. Figure 8 and Figure 9 quantify gains over DeepLabV3+ at the best epoch.

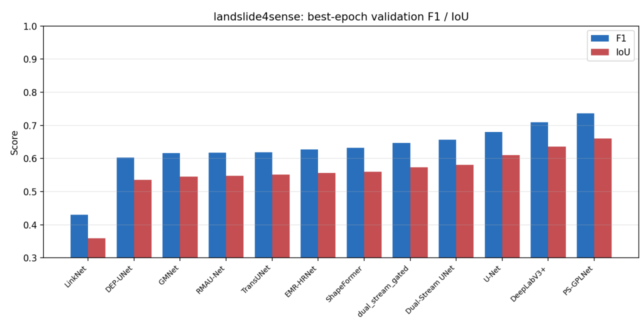


Figure 2: Landslide4Sense grouped metrics at each model's best validation epoch.

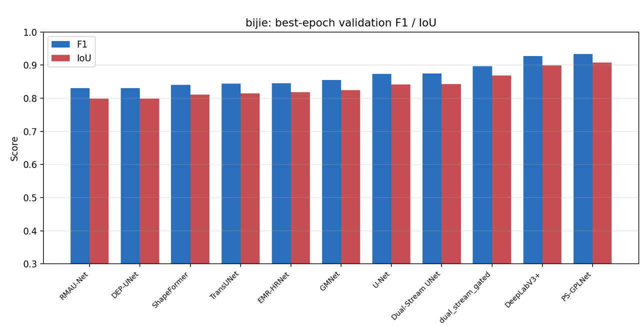


Figure 3: Bijie grouped metrics at each model's best validation epoch.

Training curves for PS-GPLNet are shown in Figure 10 and Figure 11. Precision–recall operating points for all models appear in Figure 12 and Figure 13.

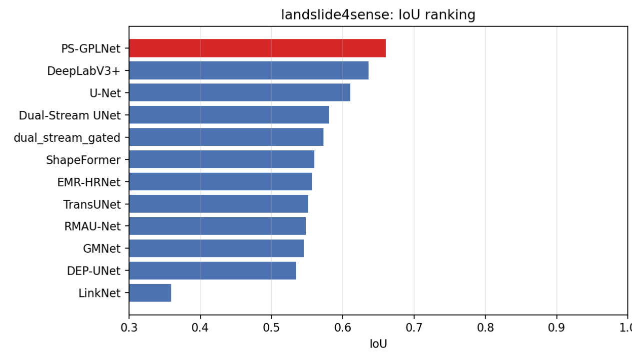


Figure 4: Landslide4Sense IoU ranking at best epoch.

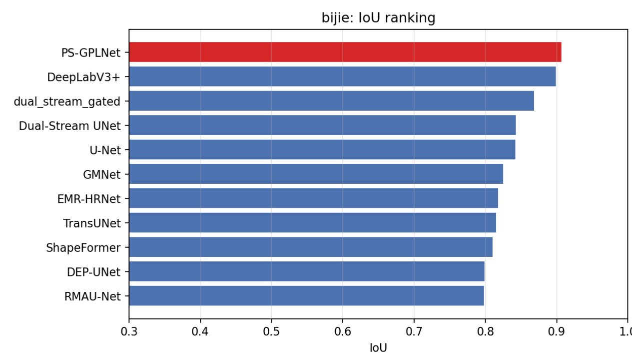


Figure 5: Bijie IoU ranking at best epoch.

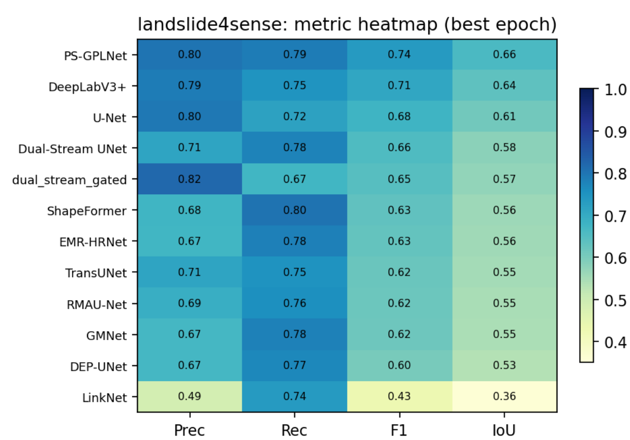


Figure 6: Landslide4Sense precision/recall/F1/IoU heatmap.

4.4 Qualitative results

Representative Bijie and Landslide4Sense patches are shown in Figure 14 and Figure 15. On Bijie, polygons are large

Table 2: Landslide4Sense best-epoch validation summary. PS-GPLNet (ours) at bottom.

Model	Acc	Prec	Rec	F1	IoU
LinkNet	0.973	0.487	0.735	0.431	0.359
DEP-UNet	0.977	0.669	0.770	0.603	0.535
GMNet	0.978	0.669	0.779	0.616	0.545
RMAU-Net	0.977	0.691	0.756	0.618	0.548
TransUNet	0.976	0.714	0.750	0.619	0.551
EMR-HRNet	0.975	0.674	0.784	0.627	0.557
ShapeFormer	0.975	0.675	0.802	0.632	0.560
Dual-Stream UNet	0.986	0.713	0.779	0.656	0.581
dual_stream_gated	0.984	0.822	0.673	0.646	0.573
U-Net	0.984	0.795	0.719	0.680	0.611
DeepLabV3+	0.984	0.789	0.749	0.710	0.636
PS-GPLNet (ours)	**0.986**	0.802	**0.788**	**0.736**	**0.660**

Table 3: Bijie best-epoch validation summary. PS-GPLNet (ours) at bottom.

Model	Acc	Prec	Rec	F1	IoU
DEP-UNet	0.982	0.850	0.934	0.831	0.799
RMAU-Net	0.981	0.859	0.924	0.830	0.798
ShapeFormer	0.984	0.892	0.899	0.840	0.810
TransUNet	0.983	0.889	0.908	0.845	0.815
EMR-HRNet	0.983	0.903	0.905	0.846	0.818
GMNet	0.983	0.900	0.909	0.855	0.825
U-Net	0.987	0.891	0.933	0.873	0.842
Dual-Stream UNet	0.986	0.902	0.928	0.875	0.843
dual_stream_gated	0.988	0.928	0.929	0.897	0.869
DeepLabV3+	0.989	0.929	0.966	0.927	0.899
PS-GPLNet (ours)	**0.991**	**0.956**	0.946	**0.933**	**0.907**

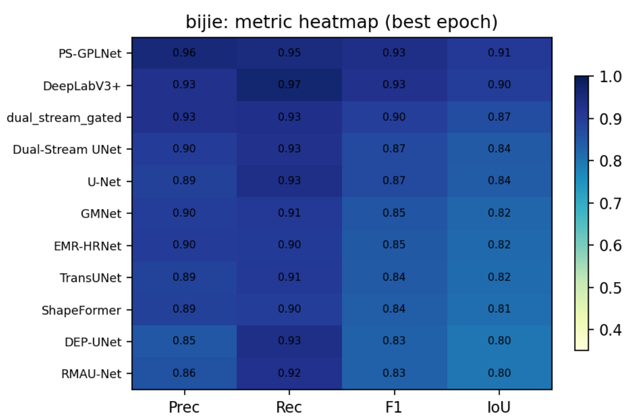


Figure 7: Bijie precision/recall/F1/IoU heatmap.

and coherent; PS-GPLNet heatmaps concentrate on scarps and debris fans while suppressing homogeneous steep slopes. On Landslide4Sense, scars are smaller and more variable across regions; the model still recovers deposit extent but confuses some spectrally similar bare soil in shadowed valleys—consistent with the precision–recall trade-off in Table 2.

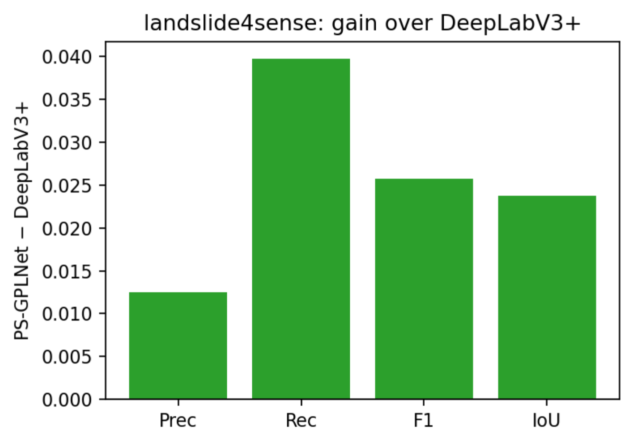


Figure 8: Landslide4Sense: PS-GPLNet minus DeepLabV3+ at best epoch.

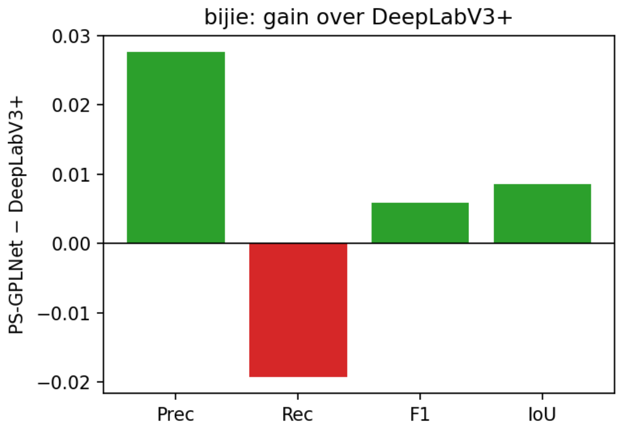


Figure 9: Bijie: PS-GPLNet minus DeepLabV3+ at best epoch.

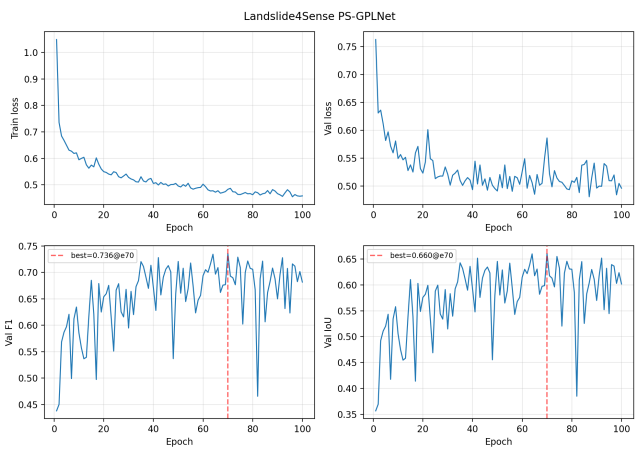


Figure 10: PS-GPLNet training dynamics on Landslide4Sense.

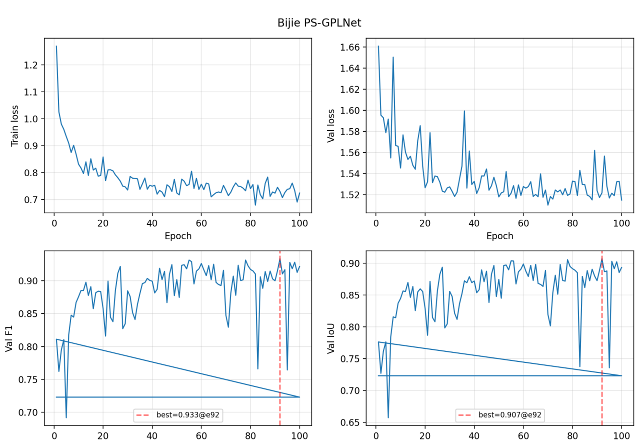


Figure 11: PS-GPLNet training dynamics on Bijie (best F1 at epoch 92).

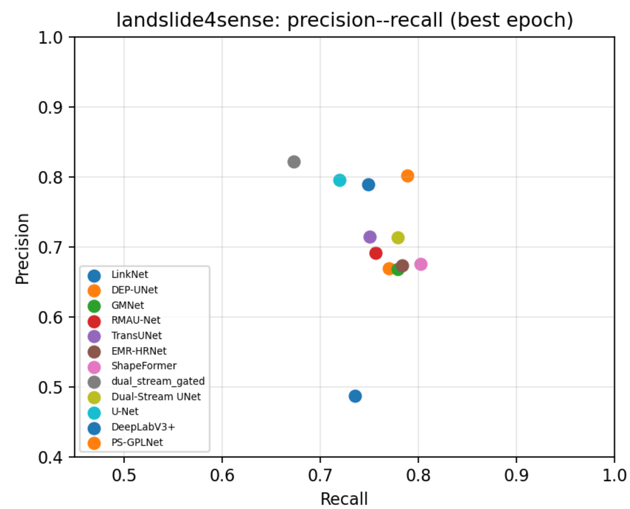


Figure 12: Precision–recall comparison on Landslide4Sense (best epoch per model).

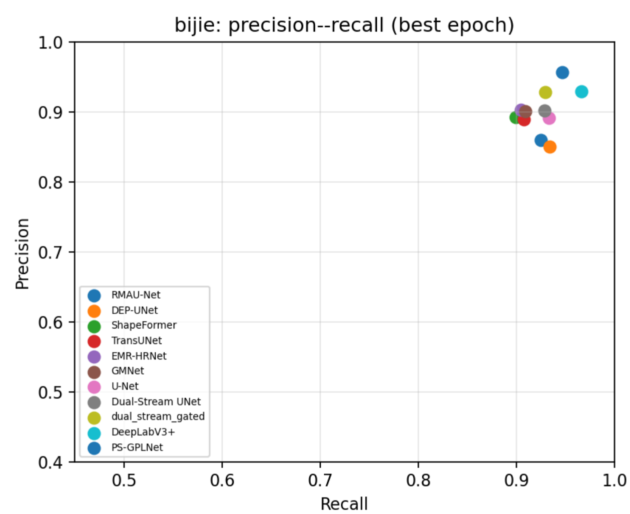
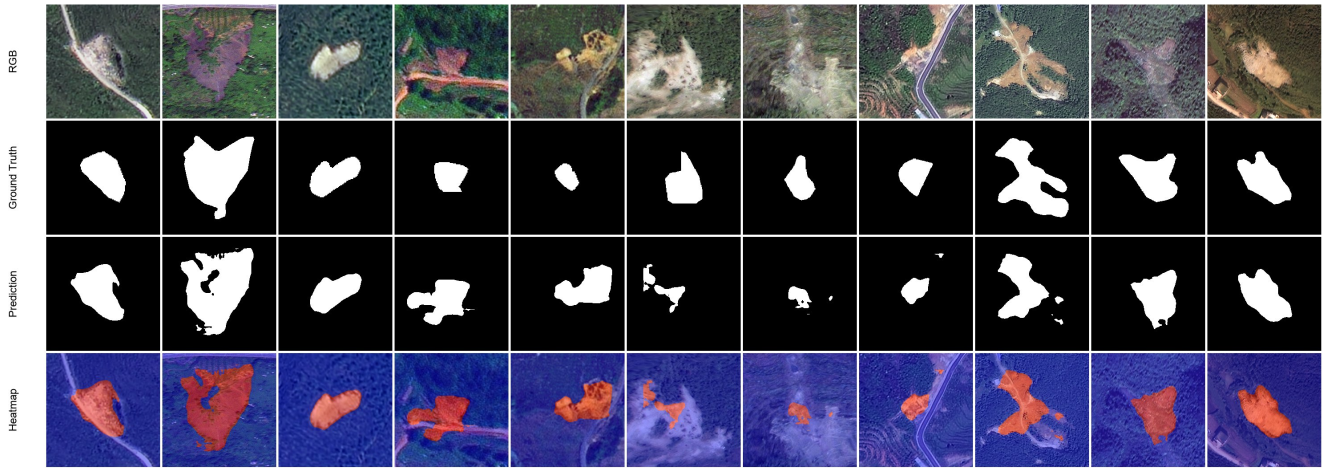


Figure 13: Precision–recall comparison on Bijie (best epoch per model).



(a) Bijie Landslide Dataset

Figure 14: Qualitative results on Bijie validation samples: RGB, ground truth, prediction, and probability heatmap.

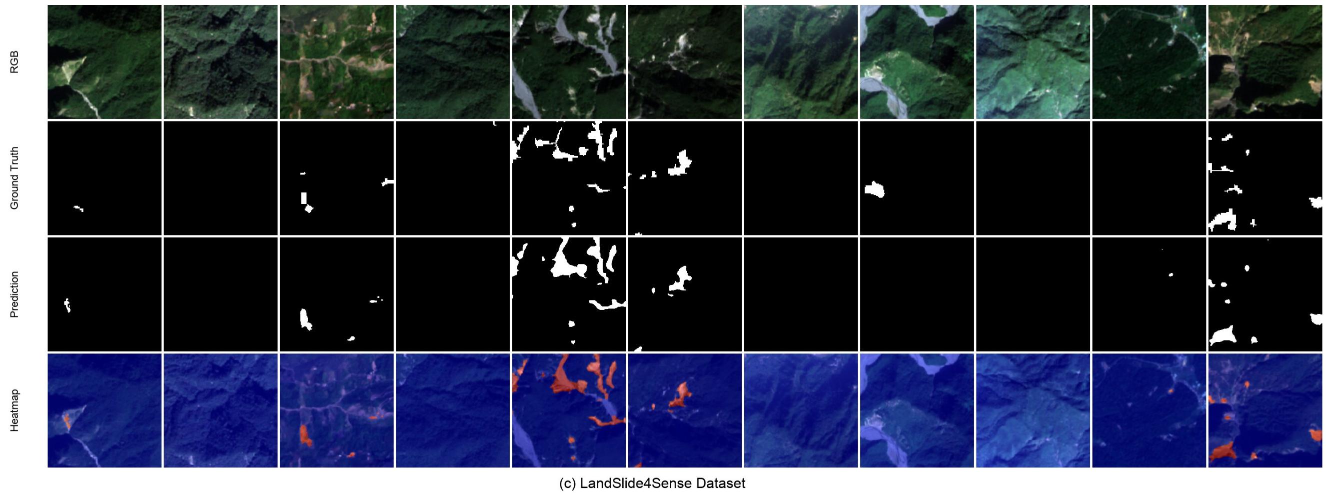


Figure 15: Qualitative results on Landslide4Sense validation samples.

5 Discussion

Novelty. Unlike susceptibility PINNs [13], PS-GPLNet produces dense masks with foundation-model context. Unlike DiGATe [10], routing is failure-energy equilibrium (MPEF), and encoding is pentastream with CMB—not two EfficientNets plus scalar GateFuse.

Limitations. Full-width EfficientNet-B4 training requires sufficient GPU memory on shared HPC nodes. Compact presets reduce width and backbone size without removing dual decoders. AUROC was not logged uniformly across baseline harness runs; we report thresholded F1/IoU for consistency.

6 Conclusion

We presented GeoPhysicsLandslideNet, a physics-closed pentastream landslide segmenter. Closing FS logic through encoding, fusion, decoding, and MPEF yields leading Bijie F1/IoU in our unified comparison and competitive Landslide4Sense scores. Future work includes ablating individual physics blocks, multi-temporal Prithvi conditioning, and coupling dynamic rainfall forcings into the moisture proxy m.

Acknowledgements

We thank CSIR-4PI for compute and research environment, and the maintainers of Landslide4Sense, Bijie, Prithvi-EO, and open baseline implementations.

References

[1] Omid Ghorbanzadeh et al. Evaluation of different machine learning methods and deep-learning convolutional neural networks for landslide detection. *Remote Sensing*, 2019.

[2] Omid Ghorbanzadeh et al. Landslide4sense: Reference benchmark data and deep learning models for landslide detection. *IEEE Transactions on Geoscience and Remote Sensing*, 2022.

[3] Shunping Ji, Dawen Yu, Chaoyong Shen, Weile Li, and Qiang Xu. Landslide detection from an open satellite imagery and digital elevation model dataset. *Landslides*, 2020.

[4] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. *MICCAI*, 2015.

[5] Liang-Chieh Chen, Yukun Zhu, George Papandreou, Florian Schroff, and Hartwig Adam. Encoder-decoder with atrous separable convolution for semantic image segmentation. In *ECCV*, 2018.

[6] Enze Xie et al. Segformer: Simple and efficient design for semantic segmentation with transformers. 2021.

[7] Jieneng Chen et al. Transunet: Transformers make strong encoders for medical image segmentation. *arXiv preprint arXiv:2102.04306*, 2021.

[8] Alexey Dosovitskiy et al. An image is worth 16x16 words: Transformers for image recognition at scale. 2021.

[9] Alexandre Mercier et al. Bifusion-net: A multi-stream fusion network for landslide detection using high-resolution remote sensing imagery and dem. *Remote Sensing*, 14(3):612, 2022.

[10] Md Islam et al. Digate: Dual-stream gated attention network for landslide segmentation. *Remote Sensing*, 2024.

[11] Nathan M Newmark. Effects of earthquakes on dams and embankments. *Géotechnique*, 1965.

[12] A W Skempton and F A DeLory. Stability of natural slopes in london clay. *Proc. 4th Int. Conf. Soil Mech.*, 1957.

[13] Ashok Dahal and Luigi Lombardo. Towards physics-informed neural networks for landslide prediction. *Engineering Geology*, 337:107852, 2024.

[14] Songlin Liu, Luqi Wang, Wengang Zhang, et al. A physics-informed data-driven model for landslide susceptibility assessment in the three gorges reservoir area. *Geoscience Frontiers*, 2023.

[15] A. Maurya et al. Landslide susceptibility mapping using physics-guided machine learning. *Natural Hazards*, 2024.

[16] George Em Karniadakis et al. Physics-informed machine learning. *Nature Reviews Physics*, 3:422–440, 2021.

[17] Daniel Restrepo et al. Physics-informed machine learning for geosciences: A review. *Reviews of Geophysics*, 2022.

[18] Biswajeet Pradhan. Landslide susceptibility assessment using gis and artificial intelligence. *Geomatics, Natural Hazards and Risk*, 2017.

[19] Johannes Reiche et al. Characterizing landslide fast-moving processes using multi-temporal sar data. *Remote Sensing of Environment*, 2013.

[20] Jonathan Long, Evan Shelhamer, and Trevor Darrell. Fully convolutional networks for semantic segmentation. In *CVPR*, 2015.

[21] Ashish Vaswani et al. Attention is all you need. 2017.

[22] Ze Liu et al. Swin transformer: Hierarchical vision transformer using shifted windows. 2021.

[23] Mingxing Tan and Quoc Le. Efficientnet: Rethinking model scaling for convolutional neural networks. 2019.

[24] Seyed Sadegh Mohseni Salehi, Deniz Erdogmus, and Ali Gholipour. Tversky loss function for image segmentation using 3d fully convolutional deep networks. 2017.

[25] Tsung-Yi Lin et al. Focal loss for dense object detection. 2017.

[26] Daniela Szwarcman et al. Prithvi-eo-2.0: A versatile multi-temporal foundation model for earth observation applications. *arXiv preprint*, 2024.

[27] Michael Jakubowski et al. Foundation models for generalist geospatial artificial intelligence. *arXiv preprint arXiv:2405.20486*, 2024.

[28] Isha Chawla et al. Geoai and foundation models in remote sensing. *IEEE Geoscience and Remote Sensing Magazine*, 2023.

[29] Neil Houlsby et al. Parameter-efficient transfer learning for nlp. 2019.